new light on the massive climate change that seems to have occurred in early Martian history. It would have photographed some of the surface of Mars with detail better than two meters across. Of course, we do not know what wonders *Mars Observer* would have uncovered. But every time we examine a world with new instruments and in vastly improved detail, a dazzling array of discoveries emerges—just as it did when Galileo turned the first telescope toward the heavens and opened the era of modern astronomy.

According to the Commission of Inquiry, the cause of the failure was probably a rupture of the fuel tank during pressurization, gases and liquids sputtering out, and the wounded spacecraft spinning wildly out of control. Perhaps it was avoidable. Perhaps it was an unlucky accident. But to keep this matter in perspective, let's consider the full range of missions to the Moon and the planets attempted by the United States and the former Soviet Union:

In the beginning, our track records were poor. Space vehicles blew up at launch, missed their targets, or failed to function when they got there. As time went on, we humans got better at interplanetary flight. There was a learning curve. The adjacent figures show these curves (based on NASA data with NASA definitions of mission success). We learned very well. Our present ability to fix spacecraft in flight is best illustrated by the *Voyager* missions described earlier.

We see that it wasn't until about its thirty-fifth launch to the Moon or the planets that the cumulative U.S. mission success rate got as high as 50 percent. The Russians took about 50 launches to get there. Averaging the shaky start and the better recent performance, we find that both the United States and Russia have a cumulative *launch* success rate of about 80 percent. But the cumulative *mission* success rate is still under 70 percent for the U.S. and under 60 percent for the U.S.S.R./*R*ussia. Equivalently, lunar and planetary missions have failed on average 30 or 40 percent of the time.

Missions to other worlds were from the beginning at the cutting edge of technology. They continue to be so today. They are designed